

HAZARDWISE

Closed-Loop Hazard Control and Verified Remediation

From Detection to Economically Provable Closure

CORE THESIS

HazardWise is designed to identify measurable hazards or losses, route corrective action, and produce evidence that the relevant condition actually changed?and what that verified closure was worth.

DETECT

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BIND

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QUANTIFY

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PRESCRIBE

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EXECUTE

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VERIFY

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CLOSE

EXECUTIVE THESIS

The product is not hazard detection. It is verified closure.

Most hazard systems are good at producing alerts, inspections, tickets, reports, or risk scores. The commercial gap appears later: **who owns the control point, what action is required, whether that action occurred, whether the physical or operational state changed, and what that closure was worth.**

SYSTEM RULE

Completion is an activity state. Closure is an evidence state.

HazardWise therefore treats a hazard as a controlled state-transition problem:



The commercial screen adds one more constraint:

COMMERCIAL RULE

Prioritize hazards where the control party, remediation payer, and economic beneficiary are the same—or where the buyer captures a material share of the value created by closure.

MEASUREMENT

Observable

The hazardous or loss state can be measured before and after intervention.

AUTHORITY

Controllable

A specific asset, process, operator, or organization has practical authority to act.

EVIDENCE

Verifiable

Execution evidence and a post-control condition can support a pass/fail closure decision.

VALUE

Economic

The payer can connect closure to recovered revenue, lower operating cost, compliance, liability reduction, asset protection, or uptime.

Scope note: This document defines a system and commercial thesis. Domain-specific thresholds, remediation methods, safety controls, and engineering judgments require validation for each deployment.

SYSTEM ARCHITECTURE

One closed loop from detected condition to verified resolution

The architecture is intentionally domain-agnostic. Hazard-specific logic changes; the control structure does not.



Layer	Function	Minimum output
Observation	Ingest sensor, meter, inspection, operational, environmental, or asset-state evidence.	Timestamped measurable condition
Binding	Associate the condition with the responsible asset/source and accountable control point.	Asset/source + responsible actor
Quantification	Prioritize by severity, exposure, consequence, urgency, or economic loss.	Risk / loss state with basis
Prescription	Route a bounded corrective action that is appropriate to the identified condition.	Action, owner, deadline, closure criterion
Execution	Capture evidence that the prescribed control was actually performed.	Execution evidence + timestamp
Verification	Measure the post-control state against the predeclared closure criterion.	Pass / fail + residual risk
Economic receipt	Where supported, calculate the measurable operational or financial effect of closure.	Recovered value / avoided loss / compliance evidence

DESIGN CONSEQUENCE

HazardWise should not reward the existence of a risk record. It should reward evidence that the relevant state moved in the intended direction.

ADMISSION CONTROL

Hard Closed-Loop Gate

A hazard enters the primary HazardWise system only if all six elements are available at economically practical cost.

#	Gate	Required evidence
1	Observable state	A measurable condition indicates hazard, exposure, vulnerability, loss, or failure.
2	Responsible control point	A specific asset, process, location, organization, or operator has practical authority to intervene.
3	Quantifiable risk	Evidence is sufficient to prioritize by severity, exposure, consequence, probability, urgency, or economic effect.
4	Prescribed control	A bounded action can reduce, contain, prevent, or mitigate the identified condition.
5	Verifiable execution	Objective evidence can establish that the prescribed action was actually performed.
6	Observable post-control condition	A measurement can show whether the relevant hazard, exposure, vulnerability, or expected consequence materially changed.

EXCLUSION RULE

If any element is unavailable at economically practical cost, exclude the hazard from the primary system.

This deliberately rejects fuzzy categories that are important but not operationally closeable. It also avoids treating personal behavior as the primary control surface when the product is intended to manage organizational hazards.

Qualifies: methane equipment leak, water-network leakage, contaminated source, constrained grid asset, industrial utility leak, pipeline integrity condition.

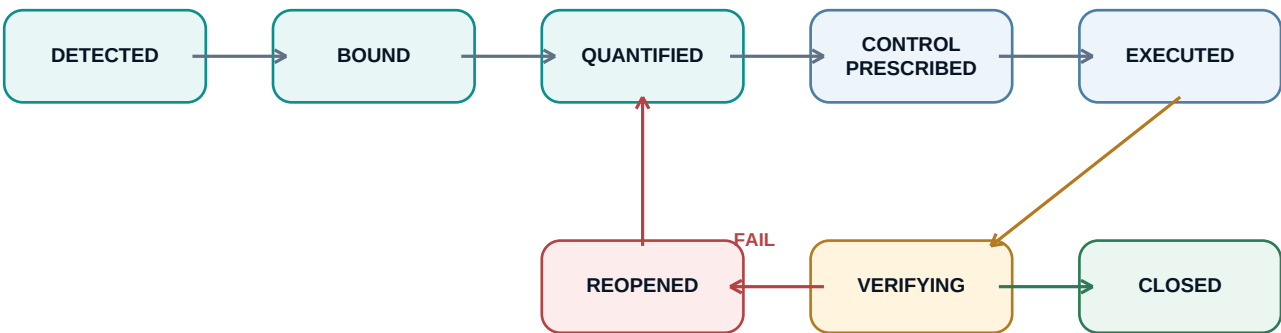
Conditional: wildfire, road-safety, and extreme-weather risks where closure must use a validated risk-state proxy rather than claiming an avoided catastrophe.

STATE CONTROL

Closure Rule: CLOSED requires evidence

HARD RULE

A hazard may enter **CLOSED** only when execution is verified and a predefined post-control measurement satisfies the applicable closure criterion.



Verification failure routes the hazard back to the earliest state that needs correction.

A completed work order, uploaded document, contractor statement, inspection record, or passage of time is **not** sufficient by itself.

Mode	Rule	Examples
Direct verification	Use when the hazardous state itself is measurable.	Methane emission rate, water balance, contaminant concentration, operating pressure / temperature.
Proxy verification	Use only when the avoided outcome cannot be observed without waiting for harm.	Fuel load, vehicle operating speed, heat exposure, warning delivery / response execution.
Verification failure	Route back to the earliest state requiring correction.	Wrong source → BOUND; inadequate control → CONTROL PRESCRIBED; recurrence → REOPENED / QUANTIFIED.

Fail-closed behavior: if required verification data are absent, ambiguous, stale, contradictory, or outside tolerance, the hazard remains open.

EVIDENCE OBJECT

Verified Closure Receipt

The core artifact is a compact, replayable record showing what was observed, what was required, what occurred, and what changed.

Receipt field	Required content
Hazard identity	Hazard ID, domain, asset/source, location, responsible actor
Pre-control state	Measurement, timestamp, method, evidence source, uncertainty/tolerance
Risk / economic exposure	Severity, exposure, expected consequence, operational loss, compliance or liability context
Prescribed control	Action, owner, due date, rule/threshold, expected state change
Execution evidence	Work record, inspection, telemetry, contractor evidence, photographs, sensor events, or equivalent
Post-control state	Repeat measurement using defined method; timestamp; comparison with baseline
Residual risk	Remaining exposure, uncertainty, monitoring interval, recurrence conditions
Closure determination	PASS / FAIL; decision rule applied; approver or automated control record
Economic effect	Recovered revenue, reduced loss, deferred capex, avoided O&M;, uptime, compliance evidence—only where supportable

KEY DISTINCTION

A hash can prove artifact identity. A Verified Closure Receipt is intended to prove the control record: the defined action occurred and the measured state satisfied the closure rule.

COMMERCIAL CLOSURE VALUE

The payer must capture enough value to care

Commercial Closure Value is not the same as social importance. A technically excellent hazard loop may still be a weak market if the remediation payer does not capture the benefit.

PAYER GATE

Reject or deprioritize hazards where the party with authority to close the loop captures too little of the economic, regulatory, operational, safety, liability, insurance, asset-protection, or uptime value.

Factor	Commercial question	Preferred condition
Control authority	Can the buyer actually direct the corrective action?	Buyer owns / operates / regulates the control point
Value capture	Does closure create value the buyer can see?	Recovered revenue, avoided cost/capex, compliance, liability, uptime
Data availability	Can the system observe the state without a bespoke science project?	Existing telemetry, meters, inspections, logs, or low-cost sensors
Closure speed	How quickly can a post-control state be verified?	Hours to months, not decades
Integration burden	How many organizations must coordinate?	One buyer or a short chain of accountable actors
Repeatability	Will the loop recur often enough to justify a system?	Portfolio of assets / recurring hazards / repeat inspections

Conceptually:

COMMERCIAL CLOSURE VALUE

Closure verifiability × captured economic value × data availability × control authority — discounted by integration cost, time-to-value, and coordination burden.

Important displacement: lead remains an unusually strong societal closed-loop opportunity, but it falls commercially when the party paying for remediation captures only a fraction of lifetime health and economic benefit. WHO nevertheless describes lead poisoning as vastly underfunded and reports large benefit-cost ratios for source-control interventions. [10]

OPPORTUNITY MAP

Primary commercial candidates

#	Hazard loop	Natural buyer	Closure evidence	Buyer value	Fit
1	Water-network leakage / non-revenue water	Water utility	Flow / pressure / water-balance normalization	Recovered billable water; lower pumping/treatment cost; deferred supply capex	EXCEPTIONAL
2	Methane leaks / super-emitters	Oil & gas operator	Repeat emission measurement below threshold	Recovered product; emissions performance; compliance	EXCEPTIONAL
3	Wastewater / sewer overflow & I/I	Wastewater utility / city	Reduced overflow / restored capacity / controlled level	Avoided capex, O&M; enforcement exposure, emergency response	EXCEPTIONAL
4	Refrigerant leakage	Food retail / cold storage	Leak rate / system state normalizes after repair	Lower refrigerant replacement and maintenance cost; compliance	VERY HIGH
5	Pipeline integrity / leak anomalies	Pipeline operator	Repair + reinspection / pressure / leak verification	Liability reduction; uptime; product loss; compliance	VERY HIGH
6	Grid congestion / thermal constraints	Utility / system operator	Verified safe hosting capacity / lower constraint	Usable capacity; reliability; deferred infrastructure	VERY HIGH
7	Industrial process / utility leaks	Manufacturer	Flow / acoustic / energy state normalizes	Immediate energy and maintenance savings	VERY HIGH
8	Utility wildfire ignition conditions	Electric utility / land operator	Treatment / equipment proxy satisfies criterion	Catastrophic-loss avoidance; reliability; regulatory exposure	HIGH

Ranking basis: fit to the closed-loop commercial architecture, not raw global harm or market-size estimation. The ranking is a synthesis, not a measured TAM forecast.

COMMERCIAL EVIDENCE

Why the top four fit

DIRECT \$ RECEIPT

1 — Water-network leakage

Loop: meter/flow/pressure anomaly → localize → quantify water/value loss → repair → verify normalized state.

Evidence anchor: a 2026 World Bank exercise across five Southern African utilities estimated 31 million m³ of annual water savings and about \$57 million/year in aggregate financial savings from NRW reduction. WHO/UNICEF GLAAS reports average NRW of 39% among reporting utilities. [1][2]

MEASURE → REPAIR → RESCAN

2 — Methane leaks

Loop: plume/emission signal → bind asset → quantify emission rate → repair → rescan → close below threshold.

Evidence anchor: IEA estimates around 70% of fossil-fuel methane emissions can be abated with existing technologies and more than 35 Mt could be avoided at no net cost at average 2025 energy prices. [3]

CAPEX + COMPLIANCE

3 — Wastewater / smart sewers

Loop: hydraulic/overflow state → locate capacity problem → route control → implement → measure overflow/capacity after.

Evidence anchor: EPA reports South Bend has saved about \$1.5M/year in O&M; and \$500M in avoided capital improvements; Louisville reported about \$117M saved versus the original plan. [4]

BORING + BANKABLE

4 — Refrigerant leakage

Loop: leak/loss signal → bind equipment → repair → verify system state / leak rate → calculate avoided replacement cost.

Evidence anchor: EPA says a typical food-retail refrigeration system leaks about one-quarter of its charge annually; GreenChill partners avoided an estimated \$134M in replacement costs in 2023. [5]

PATTERN

The strongest commercial loops do not ask the buyer to believe an abstract risk model. They produce a measurable before/after state and, often, a measurable financial receipt.

COMMERCIAL EVIDENCE

Where the architecture scales

REGULATED CONTROL LOOP

5 — Pipeline integrity

Loop: inspection/leak/threat data → risk evaluation → prescribed repair/mitigation → reinspection/validation.

Evidence anchor: PHMSA's integrity-management framework explicitly requires operators to identify, prioritize, assess, evaluate, repair, and validate pipeline integrity; 2026 guidance emphasizes risk evaluation and leak-management effectiveness. [6]

ASSET CAPACITY

6 — Grid constraints

Loop: telemetry + asset state → identify binding constraint → apply targeted enhancement → verify safe capacity/reliability.

Evidence anchor: IEA estimates grid-enhancing technologies could unlock enough capacity to connect roughly 450–700 GW of advanced-stage queued projects. [7]

FAST TIME-TO-VALUE

7 — Industrial utility leaks

Loop: acoustic/flow/energy anomaly → bind leak → repair → verify normalized consumption.

Evidence anchor: DOE Better Buildings reports compressed-air leaks can represent 20–30% of compressor output; Schneider identified more than 250 leaks associated with more than \$600,000 in savings. [8]

PROXY CLOSURE

8 — Utility wildfire risk

Loop: fuel/weather/equipment/terrain state → quantify ignition/severity risk → prescribe treatment → verify risk-state proxy after work.

Evidence anchor: UNEP reports direct wildfire response typically receives more than half of spending while planning receives less than 1%, and calls for shifting resources toward prevention/preparedness. [9]

PRODUCT BOUNDARY

HazardWise should not pretend to replace domain engineering. It should control the evidence chain: state → accountable control → executed action → verified post-control state → residual risk.

DEPLOYMENT

A bounded commercial pilot should prove closure economics

Phase	Action	Pilot KPI
0. Baseline	Select one asset class and one hazard type. Lock measurement method, evidence sources, thresholds, economic assumptions, and closure rule.	Baseline dataset; admissible evidence; closure criterion
1. Detect + bind	Run observation and anomaly logic; bind each accepted condition to a responsible asset/source and owner.	% hazards bound; false-positive / reject rate
2. Quantify + prescribe	Assign severity / loss state and route a bounded corrective action.	Time to prescription; owner acceptance; estimated \$ exposure
3. Execute + capture	Ingest work evidence without allowing completion alone to equal closure.	% actions with execution evidence; median cycle time
4. Verify	Repeat the defined measurement and apply pass/fail closure logic.	Verification pass rate; reopen rate; time to verified closure
5. Economic receipt	Calculate supported recovered value / avoided cost / capacity / compliance effect.	\$ closed; \$ per verified closure; evidence completeness

Pilot success should require more than detection accuracy:

- **Control-point binding rate:** accepted hazards have an accountable asset/source and actor.
- **False-closure rate:** closed items that later fail verification or recur.
- **Median detect-to-verified-close time:** operational cycle time.
- **Economic evidence rate:** share of closures with supportable financial or operational value.
- **Reopen discipline:** verification failures route back to the correct earlier state rather than being administratively suppressed.

GO / NO-GO COMMERCIAL TEST

Does verified closure create enough measurable value that the buyer would rationally pay to run the loop again across the rest of its asset portfolio?

CONCLUSION

The commercial thesis

HAZARDWISE

Designed to turn a measurable hazardous or loss state into an accountable, verifiable control loop—and to preserve evidence of what changed and what that closure was worth.

The highest-fit markets share the same structure:

observable physical state → identifiable asset/source → accountable control point → bounded corrective action → measurable post-control state → economic or regulatory receipt

That is the core distinction: HazardWise is not merely a detection or reporting layer. The system thesis is a controlled record from observed condition through verified resolution, with reopen behavior when the evidence does not support closure.

SELECTED EVIDENCE ANCHORS

- [1] World Bank (2026) — [Fix the Leaks: How Water Utilities Build Financial Sustainability](#)
- [2] WHO / UNICEF / UN-Water (2026) — [GLAAS 2025 key findings](#)
- [3] IEA (2026) — [Global Methane Tracker 2026: Key findings](#)
- [4] U.S. EPA (2026) — [Smart Sewers](#)
- [5] U.S. EPA — [GreenChill Store Certification / Partnership Accomplishments](#)
- [6] PHMSA (2026) — [Guidance for Enhancing the Effectiveness of Distribution Integrity Management Programs](#)
- [7] IEA (2026) — [Electricity 2026: Grids](#)
- [8] U.S. DOE Better Buildings — [Schneider Electric compressed-air leak detection and repair](#)
- [9] UNEP — [Spreading Like Wildfire / prevention and preparedness spending](#)
- [10] WHO (2026) — [Technical package on lead poisoning prevention: overview](#)

Evidence note. Figures above are evidence anchors for architecture/commercial fit, not a market-size model. Public sources were checked for this frozen v1.0. Specific deployment economics must be validated against the buyer's assets, tariffs, operating data, engineering constraints, and applicable regulatory requirements.